

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Chromosomes and their disorders

By
Dr.Noha Issa

*Ass. professor and consultant of medical
genetics*

*Faculty of medicine
KAU*

Objectives

- ✓ Classify chromosomal abnormalities
- ✓ Discuss the abnormalities on a given karyotype for a common condition
- ✓ Outline the clinical implications of abnormal karyotype



Chromosome Abnormalities

Types of chromosomal anomalies

Abnormalities of chromosomes either:

- ✓ Numerical chromosomal abnormalities
- ✓ Structural chromosomal abnormalities
- ✓ Different cell lines(mixoploidy): mosaicism

Numerical chromosomal abnormalities

Change in chromosome number involve:

- Polyploidy which is the addition of complete haploid set of chromosomes.
- Aneuploidy which is the addition or loss of one or rarely two chromosomes

Polyploidy

The basic set of chromosomes is the haploid set (n) which is the number of chromosomes present in the gametes $n=23$.

The diploid set ($2n$) is the number of chromosomes present in the normal somatic cells $2n= 46$

Polyploidy

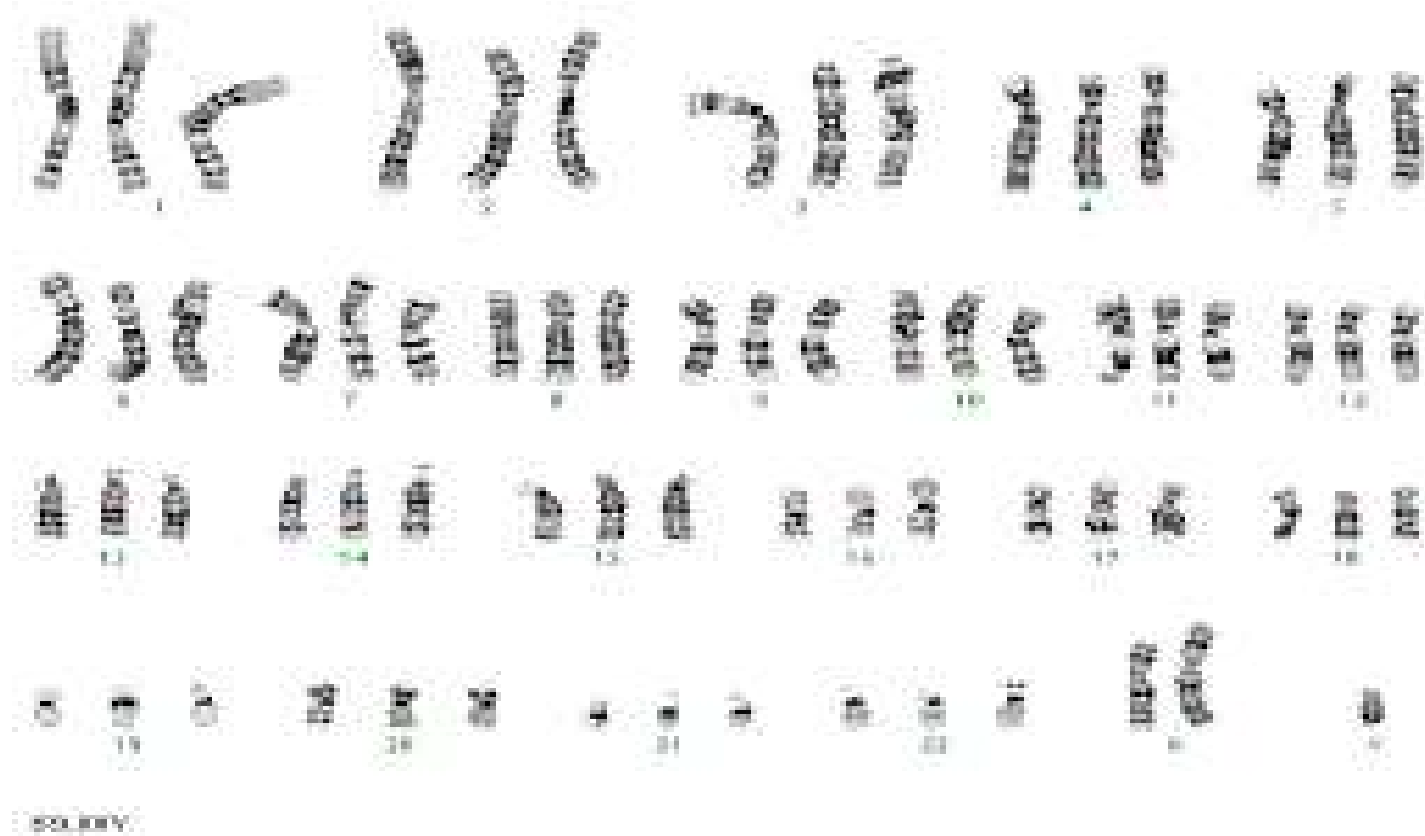
- ❖ The karyotype with any multiple of the basic haploid (n) other than the normal diploid ($2n$)
- ❖ Two polyploidy condition occur in humans:
 - ✓ Triploidy ($3n$) with 69 chromosomes
 - ✓ Tetraploidy ($4n$) with 92 chromosomes

Polyploidy

- ❖ Triploidy most frequently result from:
 - ✓ fertilization by two sperms or
 - ✓ fertilization of abnormal diploid sex cell (ova $2n$) with a normal sex cell (sperm n) or vice versa e.g Retention of polar body

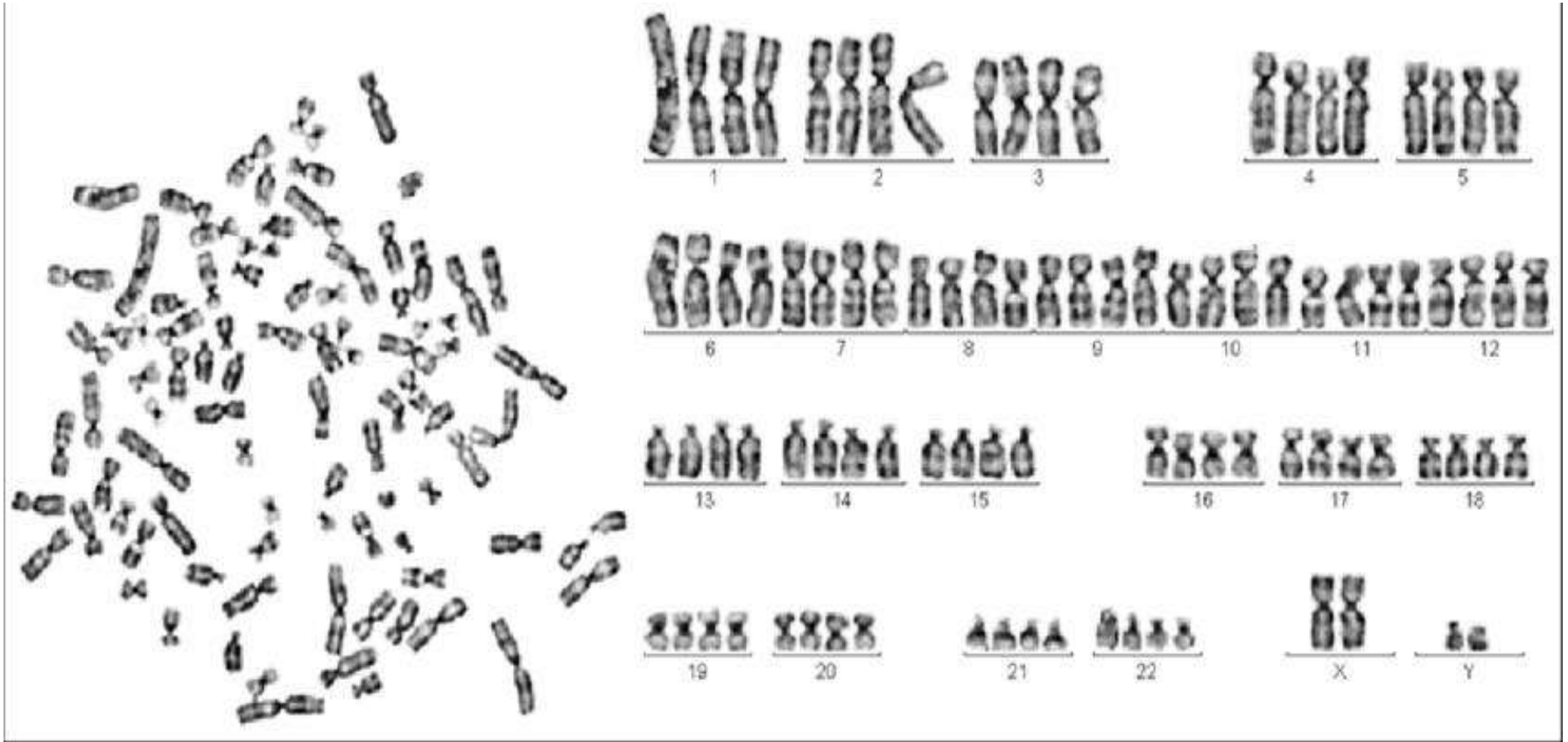
- ❖ Tetraploidy could result from the failure of the zygote to divide after it replicate its chromosomes.

Numerical chromosomal abnormalities



69,XXY

Numerical chromosomal abnormalities



92,XXYY

Aneuploidy

is the **most common** and clinically significant type of human chromosome disorder, occurring in **3-4%** of all clinically recognized pregnancies.

Most **aneuploid** patients have either **trisomy** (3 instead 2) or less often **monosomy** (1 instead of 2). Both could have severe phenotypic consequences.

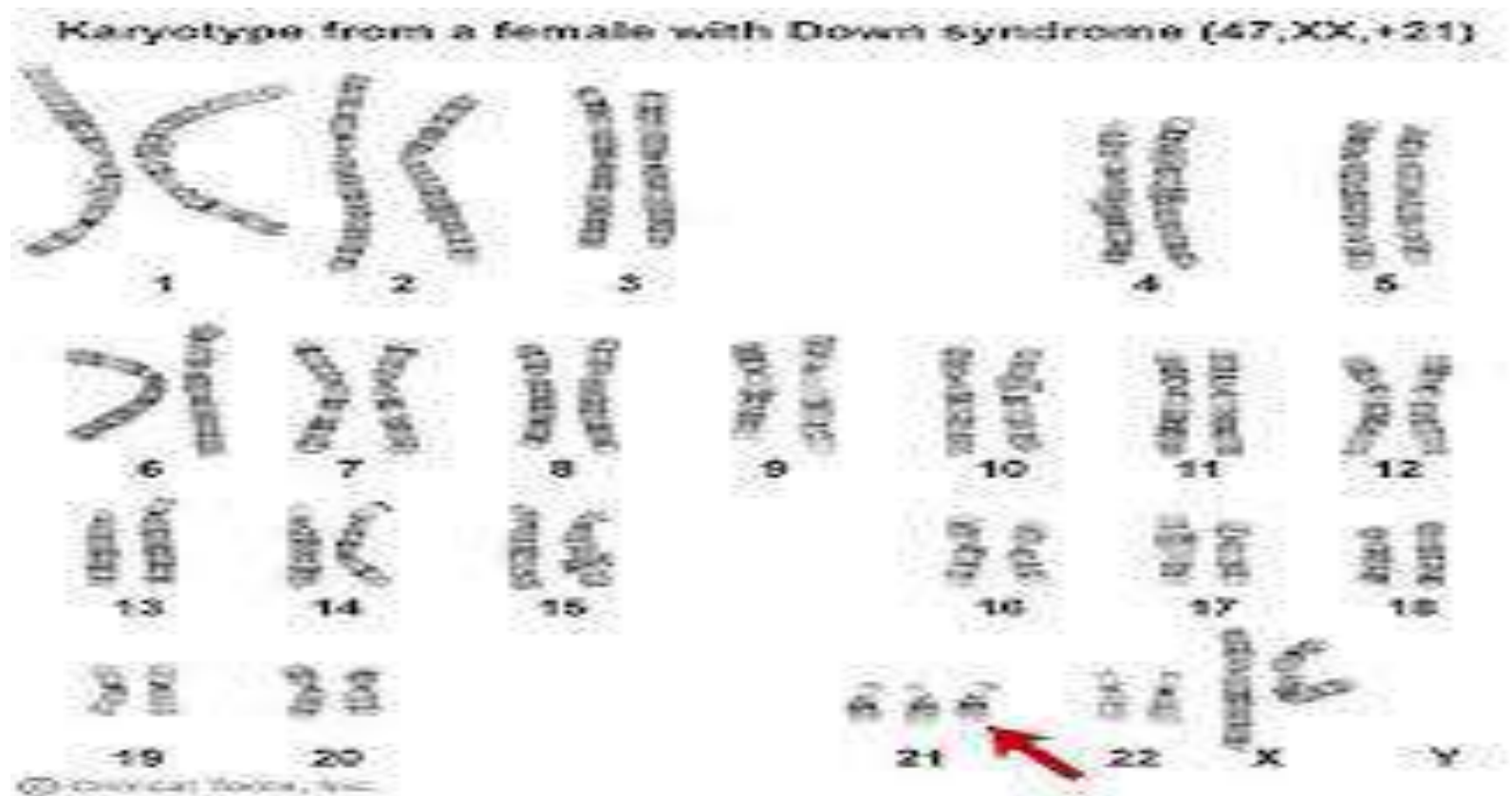
Numerical Chromosome Abnormalities

➤ **Trisomy** for a whole chromosome is rarely compatible with life except :

trisomy 21, trisomy 13, trisomy 18, triple X female, Klinefelter syndrome.

➤ **Monosomy** for an entire chromosome is almost always lethal, an important exception is **Turner syndrome**.

47,XX,+21 Standard Down Syndrome



Clinical characteristics

- **MR**
- **Craniofacial** Brachycephaly, a flat occiput, microcephaly,







- Congenital heart disease, e.g. ventricular septal defect.
- Duodenal atresia, imperforate anus
- Umbilical hernia, and cryptorchidism.
- Liability to autoimmune disorders, e.g. Thyroiditis
- Increase risk of leukemia (20 times than normal)

Down Syndrome



single palmer crease, short fifth finger that curves inward

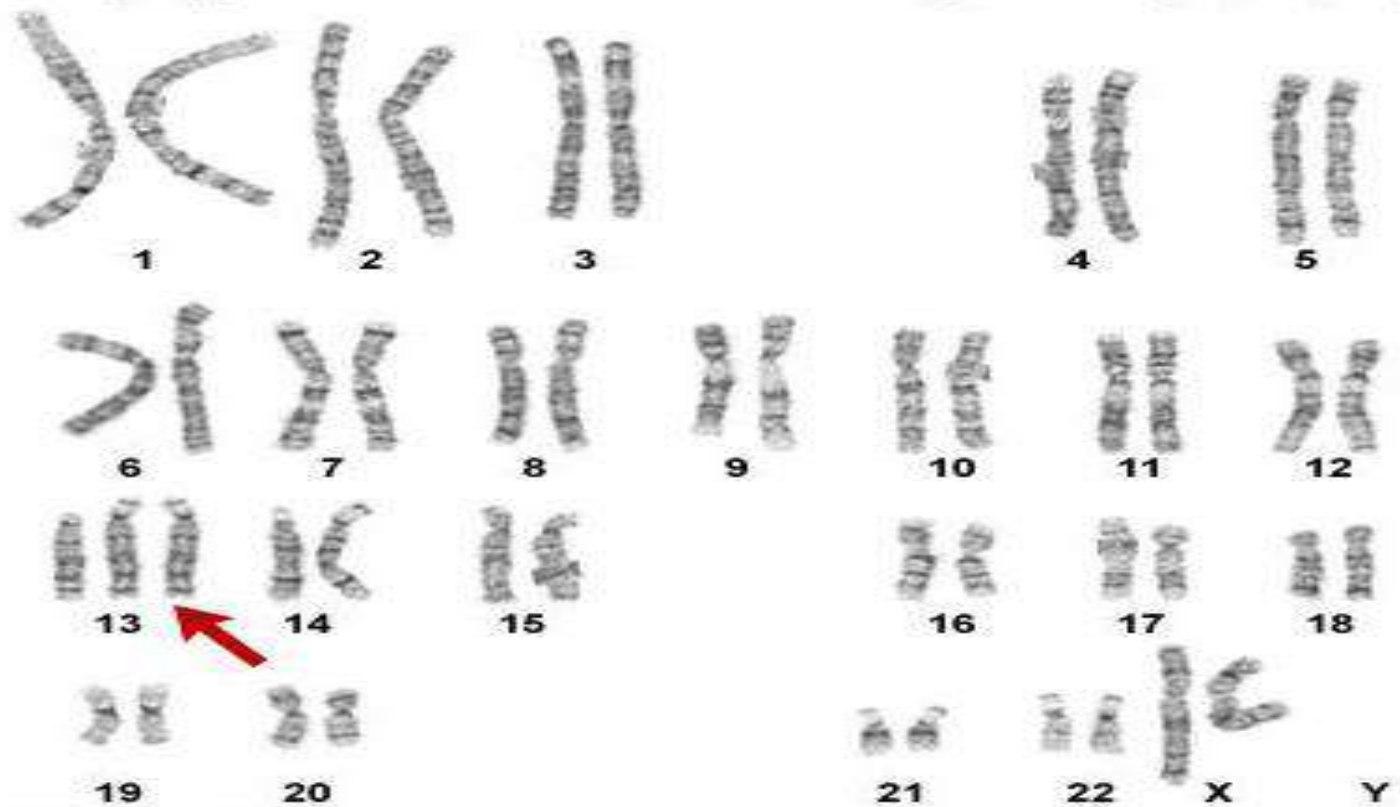


widely separated first and second toes and increased skin creases

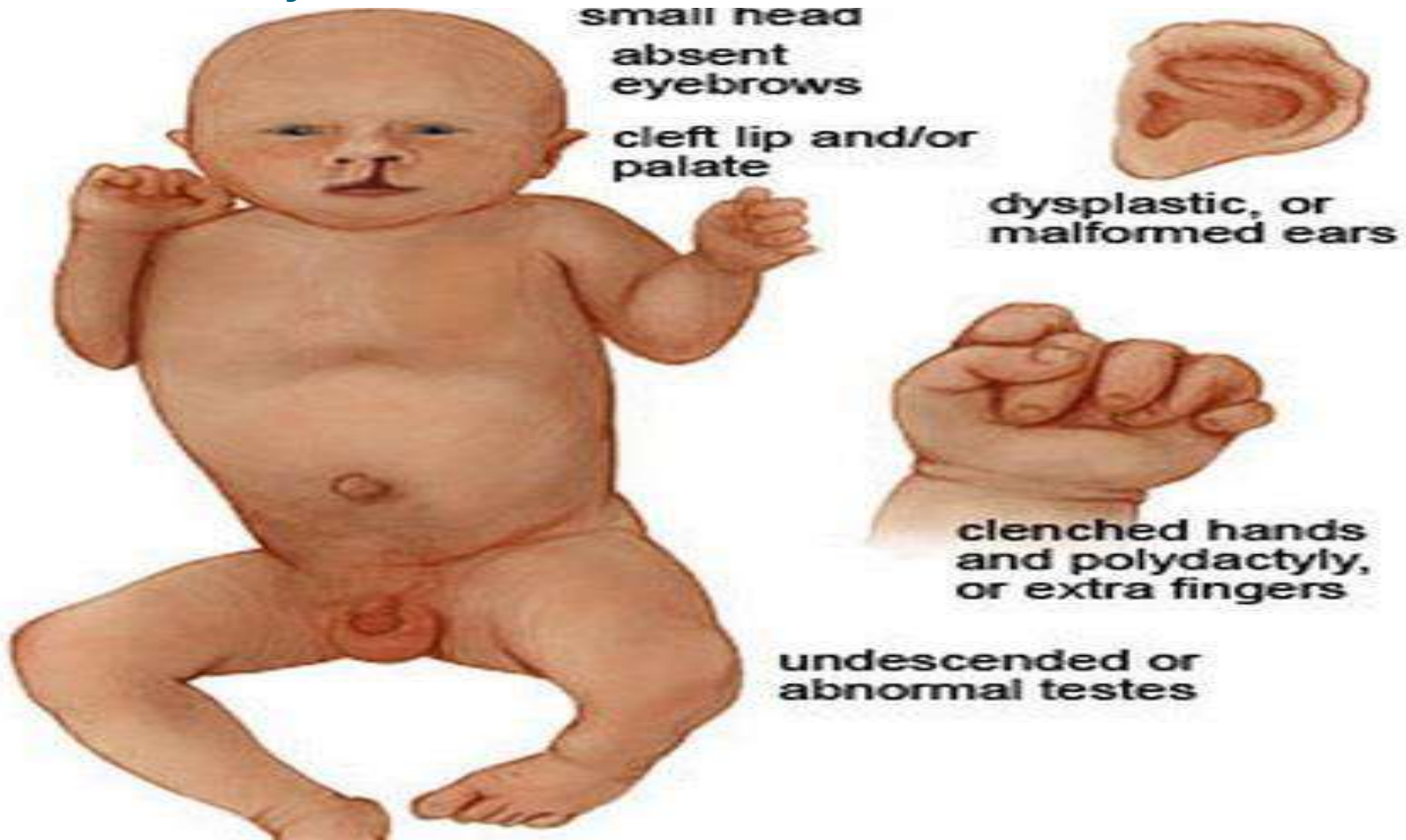
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47,XX,+13 Patau Syndrome

Karyotype From a Female With Patau syndrome (47,XX,+13)



47,XY,+13 Patau Syndrome



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Patau syndrome

47,XY,+13

Mental and growth retardation

Hypotonia, Holoprosencephaly

Microcephaly, sloping forehead, scalp defect, forebrain defect

Eye: microphthalmia, anophthalmia
and colobomata of iris

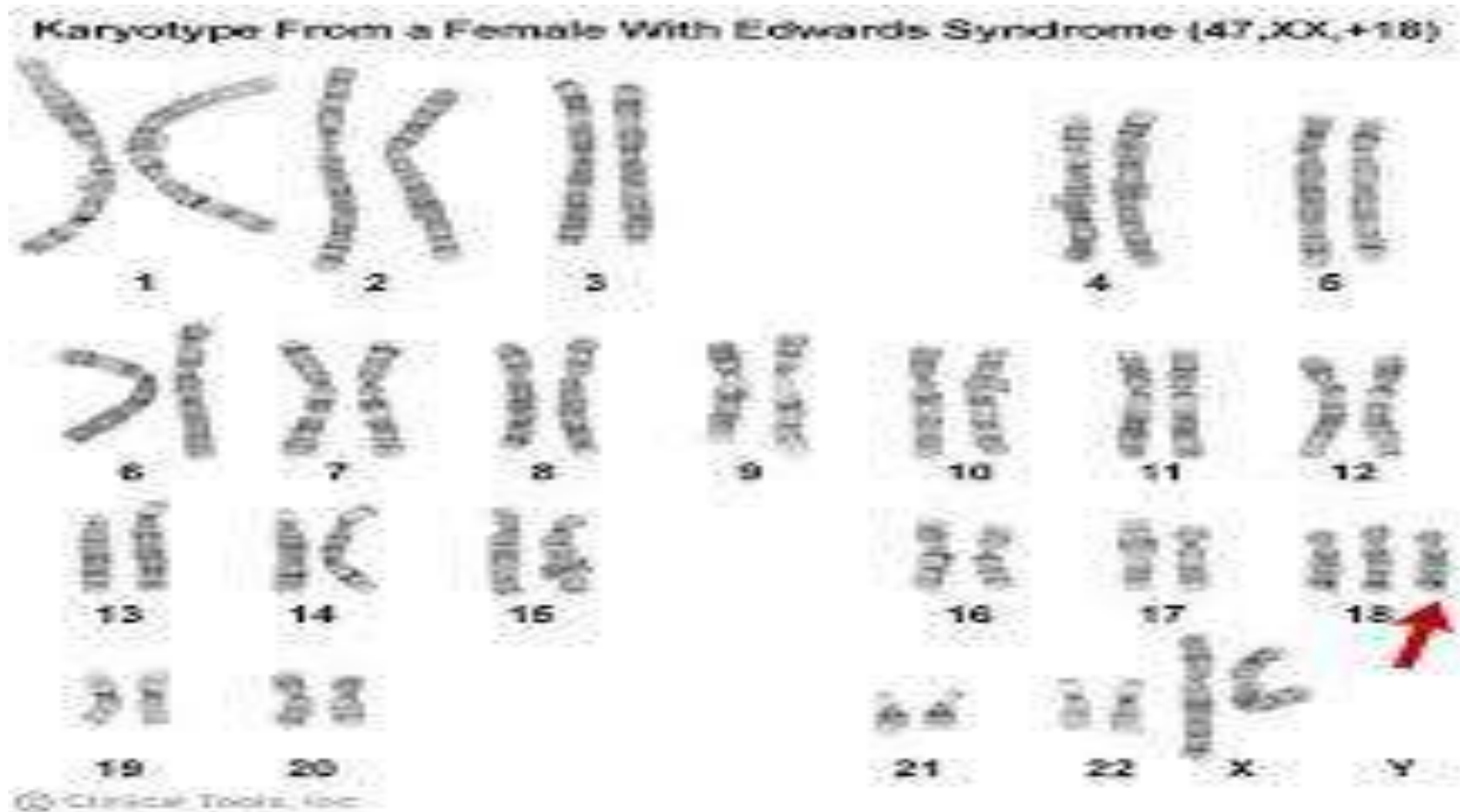
cleft lip and / or palate

Polydactyly(postaxial)

Heart: VSD, ASD, PDA



47,XX,+18 Edward Syndrome



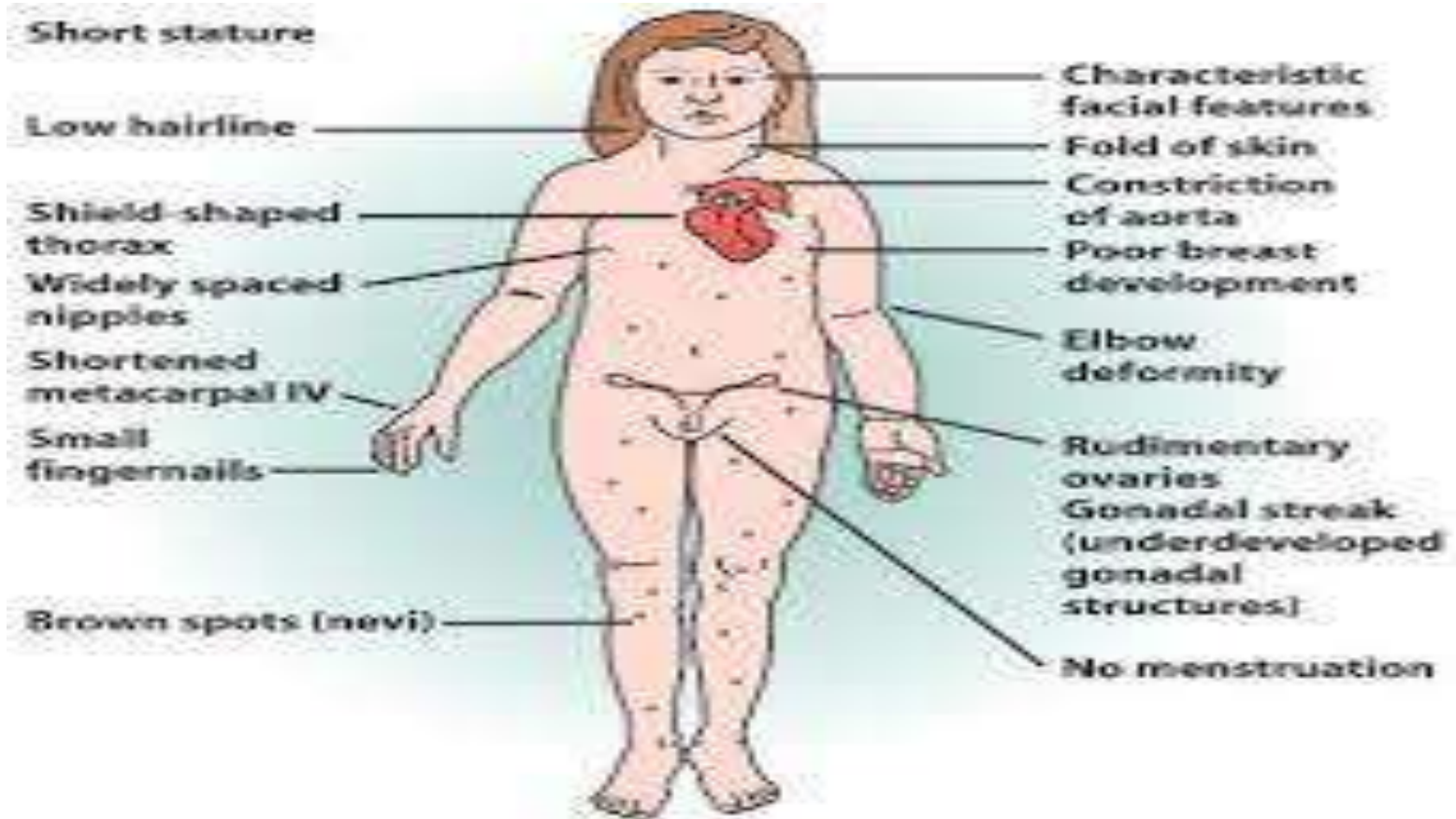
47,XY,+18 Edward Syndrome



45,X Turner Syndrome



Turner Syndrome

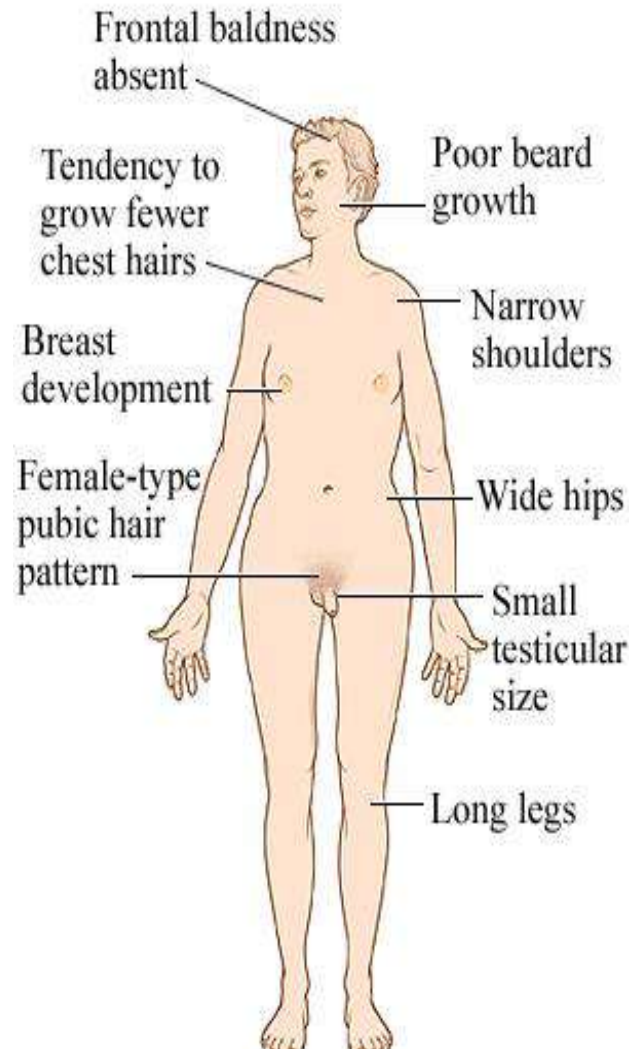


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47,XXY Klinefelter Syndrome



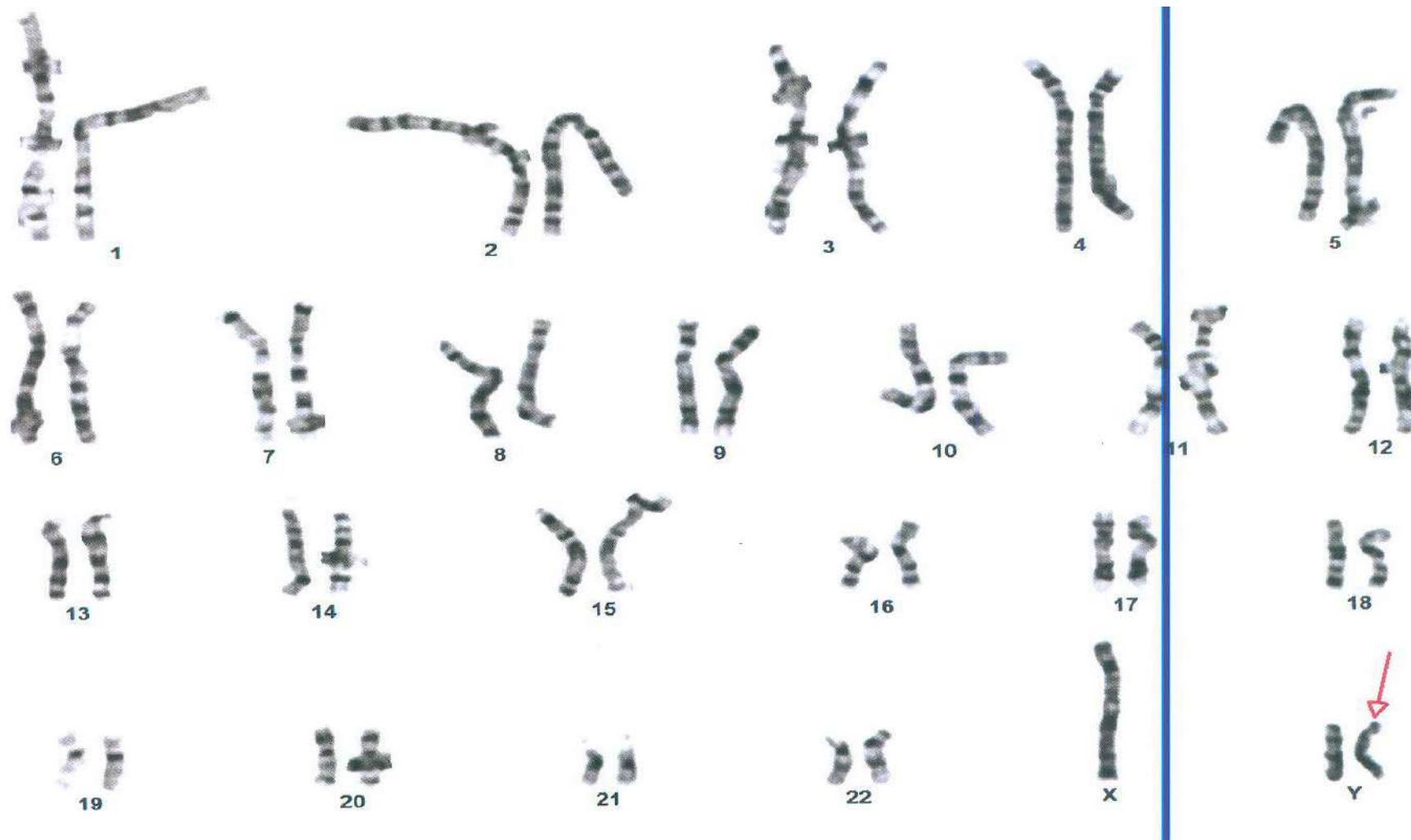
Klinefelter Syndrome



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47,XYY





Numerical Chromosome Abnormalities

The causes of aneuploidy are not well understood, but the most common chromosomal mechanism is meiotic nondisjunction. This refers to the failure of the chromosomes to disjoin properly during one of the two meiotic divisions.



Numerical Chromosome Abnormalities

Nondisjunction can occur in a **mitotic** division after the formation of a zygote. If this happens at an early cleavage division, **mosaicism** may result.

Numerical Chromosome Abnormalities

Mosaicism

- Two or more chromosome complement are present in an individual
- Mosaicism may be either numerical (most common type) or structural.

It results from:

Post-zygotic non-disjunction: both sister chromatids go to one pole--- $2n+1$ / $2n-1$

Anaphase lag: one sister chromatid lag and lost --- $2n$ / $2n-1$

Abnormalities of Chromosome Structure

It results from chromosome breakage followed by reconstitution in an abnormal combination.

These may be **balanced** if the chromosome set has the normal complement of chromosomal material, or **unbalanced** if there is additional or missing material.

Abnormalities of Chromosome Structure

Balanced	Unbalanced
The chromosome set has the normal complement ,no loss or gain	There is loss or gain of chromosomal material
Don't usually have phenotypic effect But they can pose a threat to the offspring	The phenotype is likely to be abnormal
▪ Insertion ▪ Inversion ▪ Translocations: Reciprocal or robertsonian	▪ Deletions ▪ Duplications ▪ Marker chromosomes ▪ Ring chromosomes ▪ Isochromosomes ▪ Fragile site

Symbols Used in Describing a Karyotype

Term	explanation	example
p	Short arm	
q	Long arm	
cen	centromere	
del	deletion	46,XX,del(1)(q21)
dup	duplication	46,XY,dup(13)(q14)
i	isochromosome	46,X,i(X)(q10)
r	Ring chromosome	46,XX,r(21)
t	translocation	46,XY,t(2;4)(q21;q21)
rob	Robertsonian	46,XY,rob(14;21)(q10;q10),+21
inv	inversion	46,XX,inv(9)(p12q12)
der	derivatives	46,XY,der(14;21)(q10;q10),+21
mar	marker	47,XY,+mar
ins	insertion	

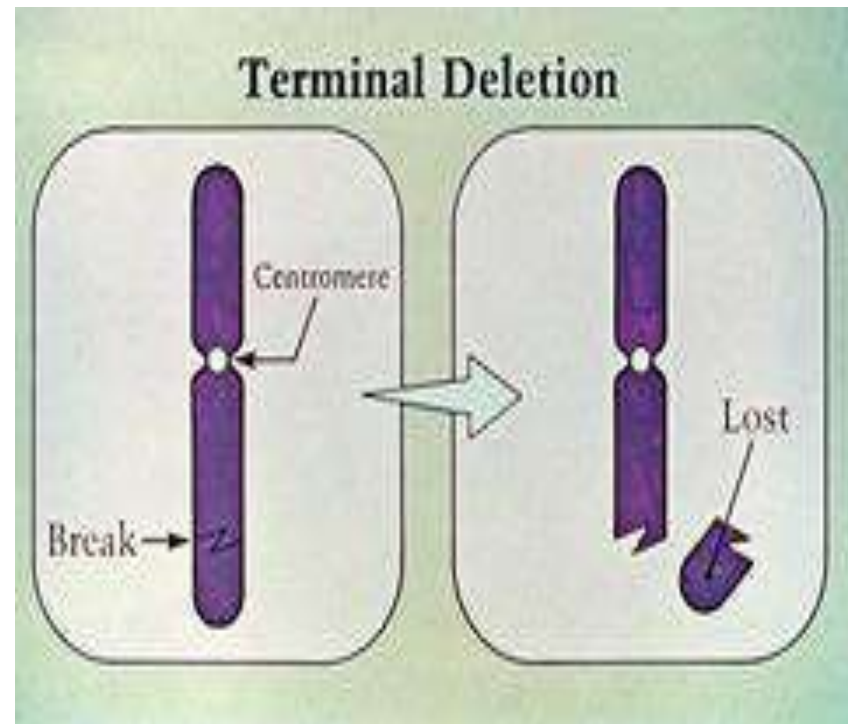
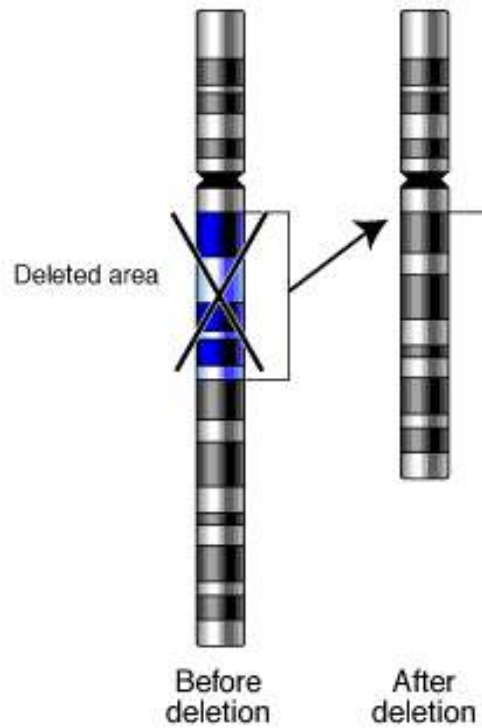


Unbalanced Rearrangements

Deletions

They involve loss of a chromosome segment resulting in chromosome imbalance.

A deletion may be **terminal or interstitial**.



Deletion

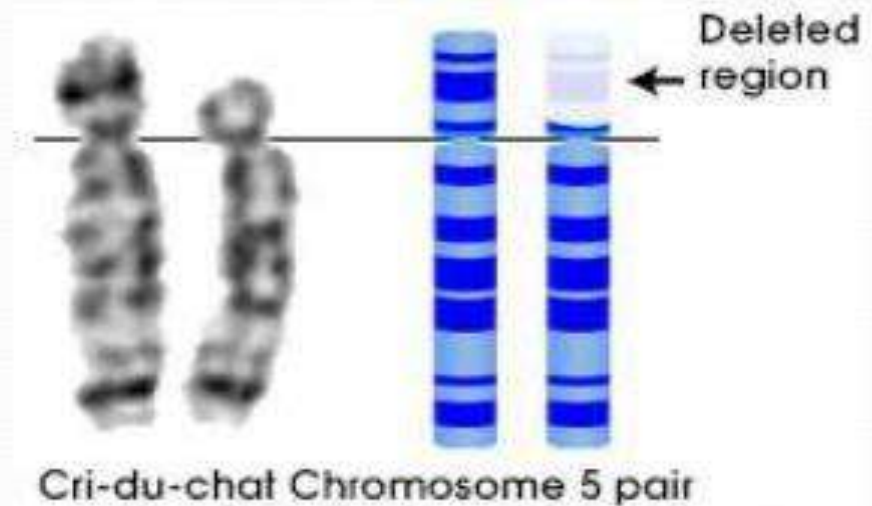
This Involves loss of a part of chromosome.

1. Terminal deletion.
2. Interstitial deletion.

1. Terminal deletion.

It involves a single break and the terminal part of the chromosome is lost.

e.g. cri-du-chat syndrome

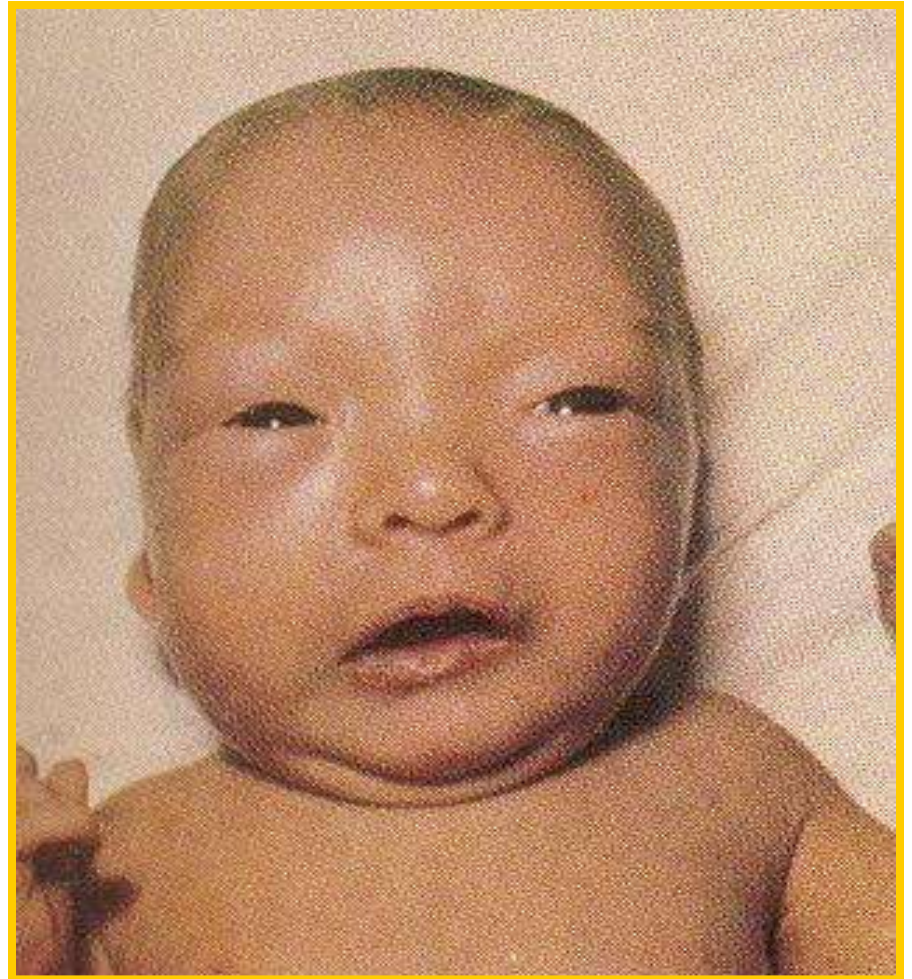


46,XX,del(5)(p15) Cri-du-chat syndrome or (5p- syndrome)



Deletion 5p15

- Cat like cry (hypoplasia of larynx)
- Moon face, hypertolerism
- Microcephaly , micrognathia
- Mental and motor delay



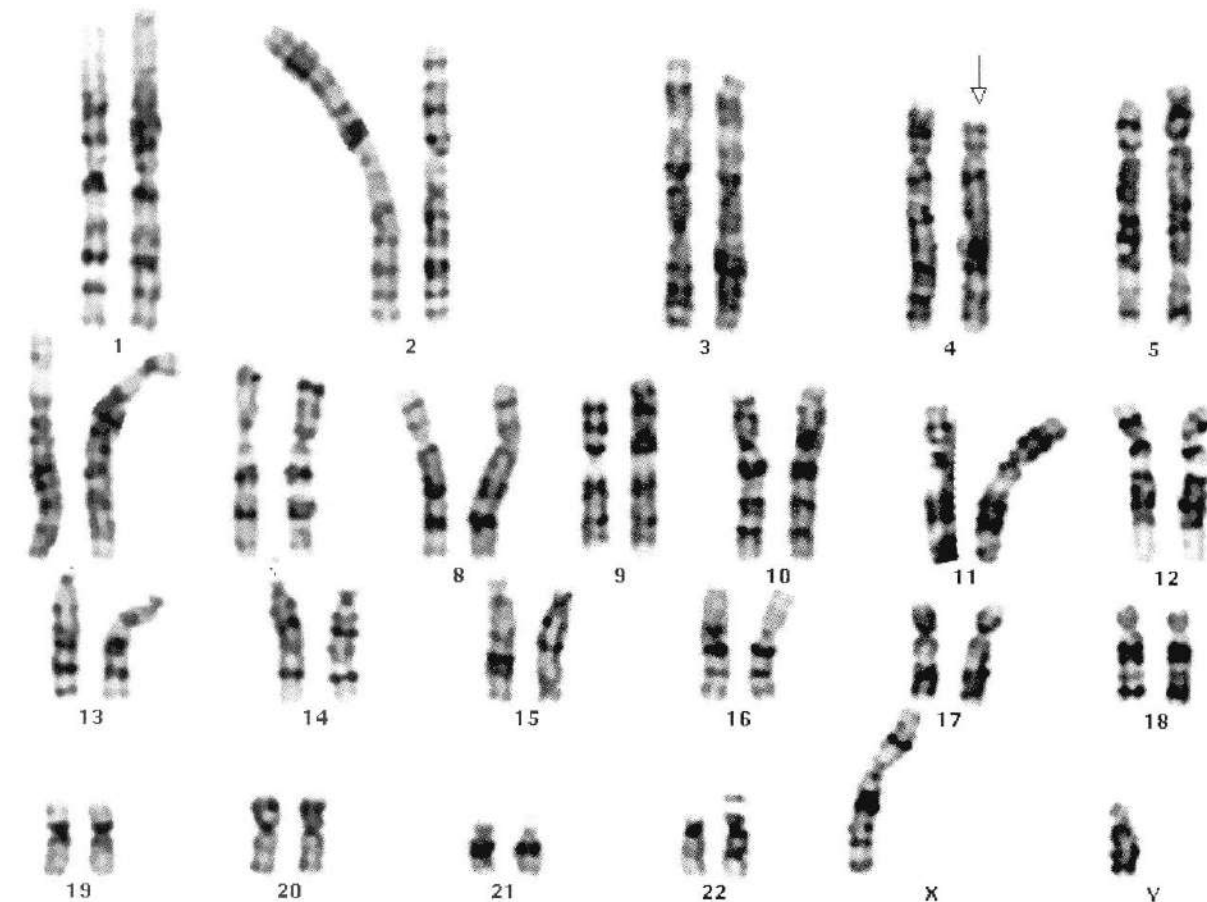
46,XY,del(4)(p16)

46,XY,del(4)(p16)

Wolf-Hirschhorn syndrome

or

4p- syndrome



46,X,del(X)(q22.1)

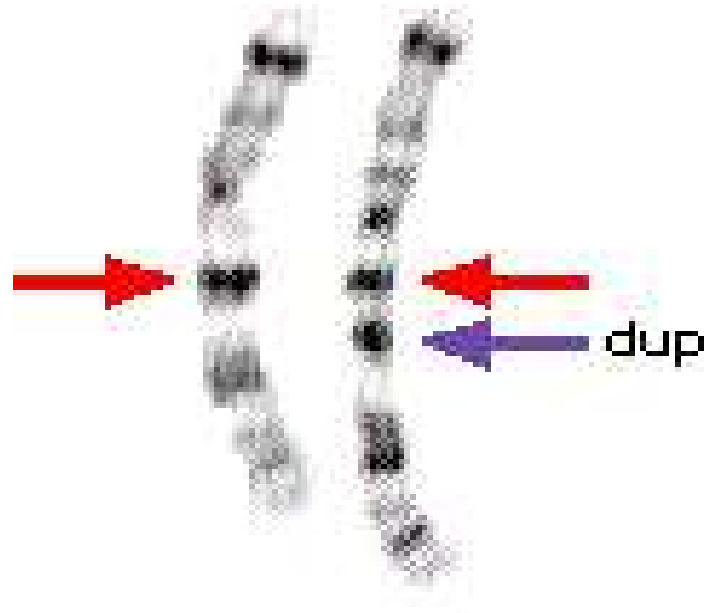
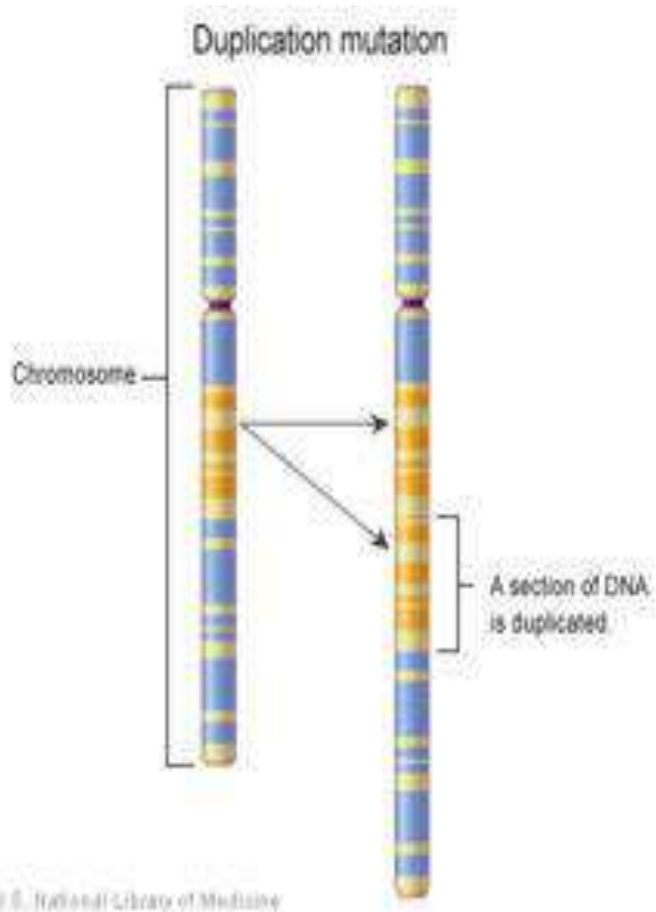




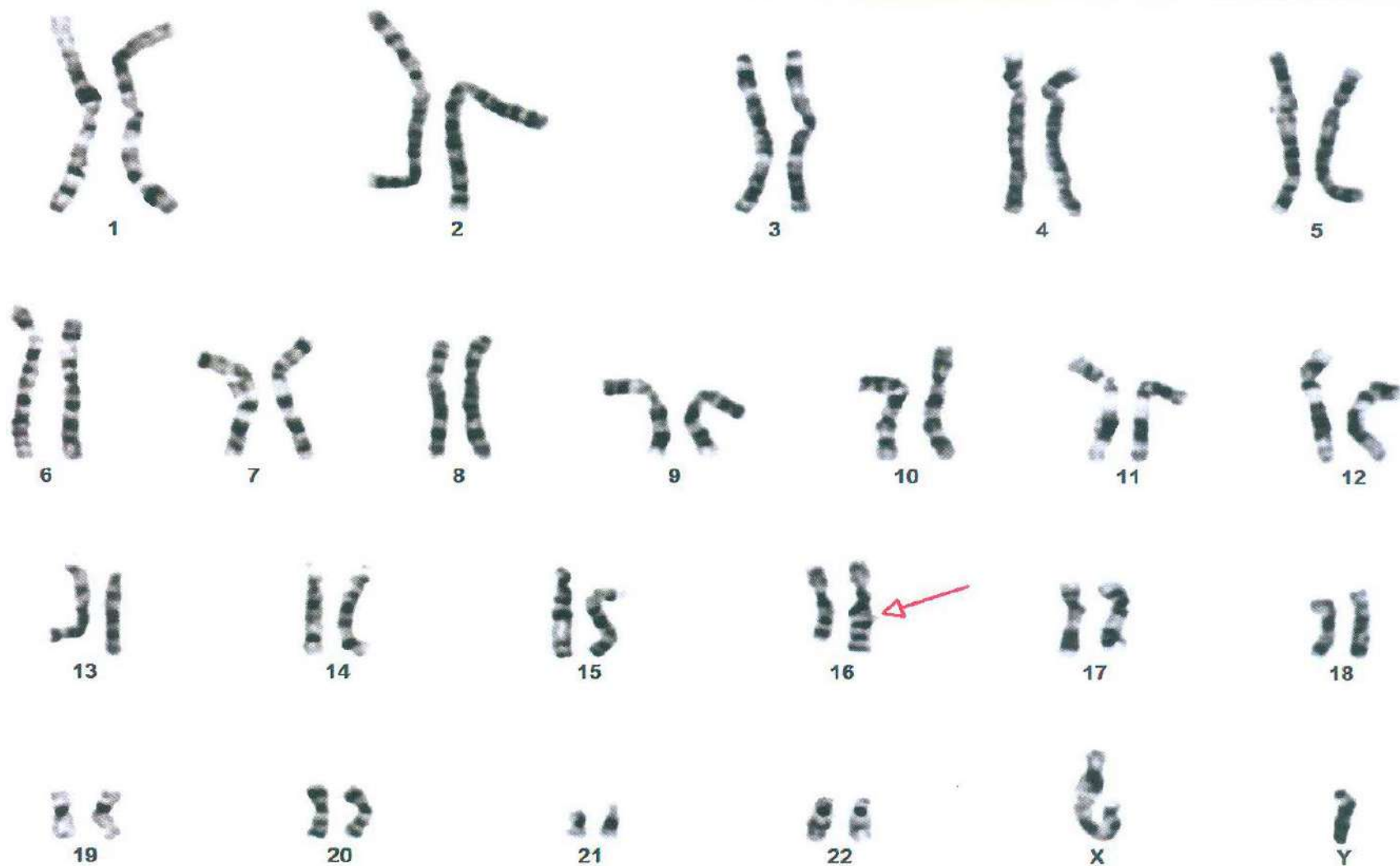
Duplications

They are less harmful than deletions. Due to partial trisomy, a chromosomal imbalance occurs leading to phenotypic abnormalities.

dup(7)(q21.1q21.3)

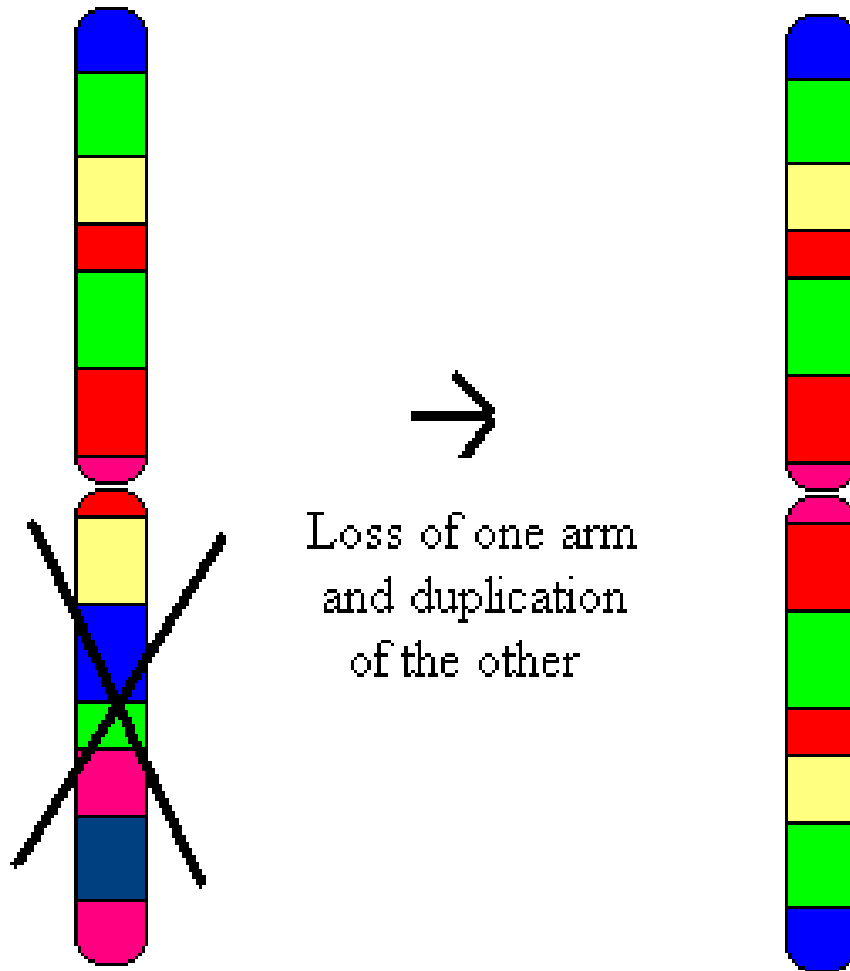


46,XY,dup(16)(q12q22)



Isochromosome

- Is a chromosome in which one arm is missing and the other duplicated in a mirror image fashion.
- Thus a person with 46, chromosomes and an isochromosome has a **single copy of one arm** and a **triple copy of the other** . The most common isochromosome is that for the **X chromosome** in some cases of **turner syndrome**.



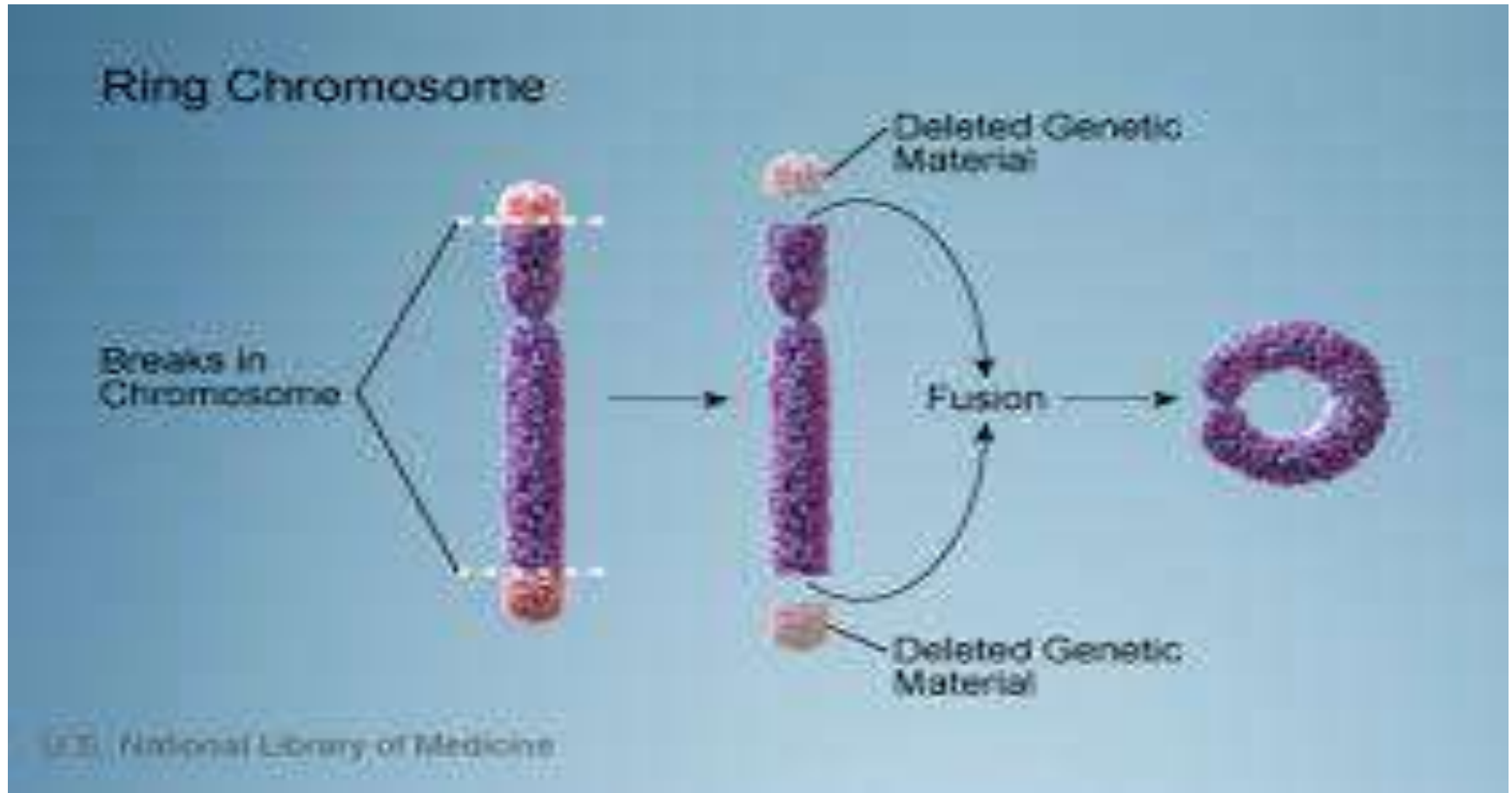
46,X,i(X)(q10) Turner syndrome



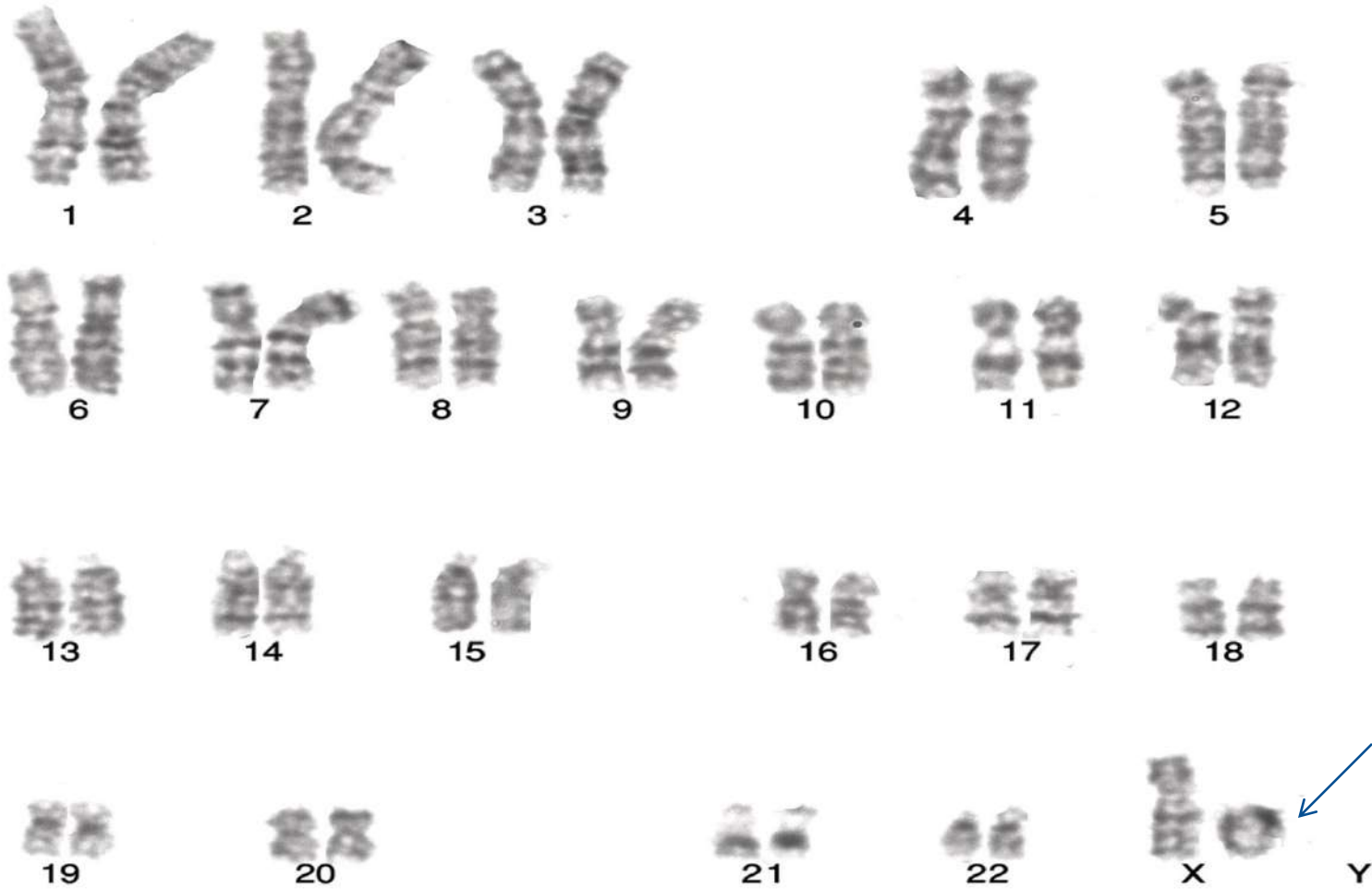
Ring chromosome

- Formed when a chromosome undergoes two breaks and the broken ends of the chromosome reunite in a ring structure.
- If the centromere is within the ring, a ring chromosome would be expected to be mitotically stable.

Ring chromosome



46,X,r(X) Turner syndrome



Chromosomal abnormalities in turner syndrome

- Numerical:

- 45,X

- Structural chromosomal abnormalities :

- E.g. : 46,X,r(X)

46,X,i(X)(q10)

46,X,del(X)(q12)

- Mosaic:

- E.g: 45,X[20]/46,XX[10]

45,X[24]/46,X,i(X)(q10)[6]

etc.



Balanced Rearrangements

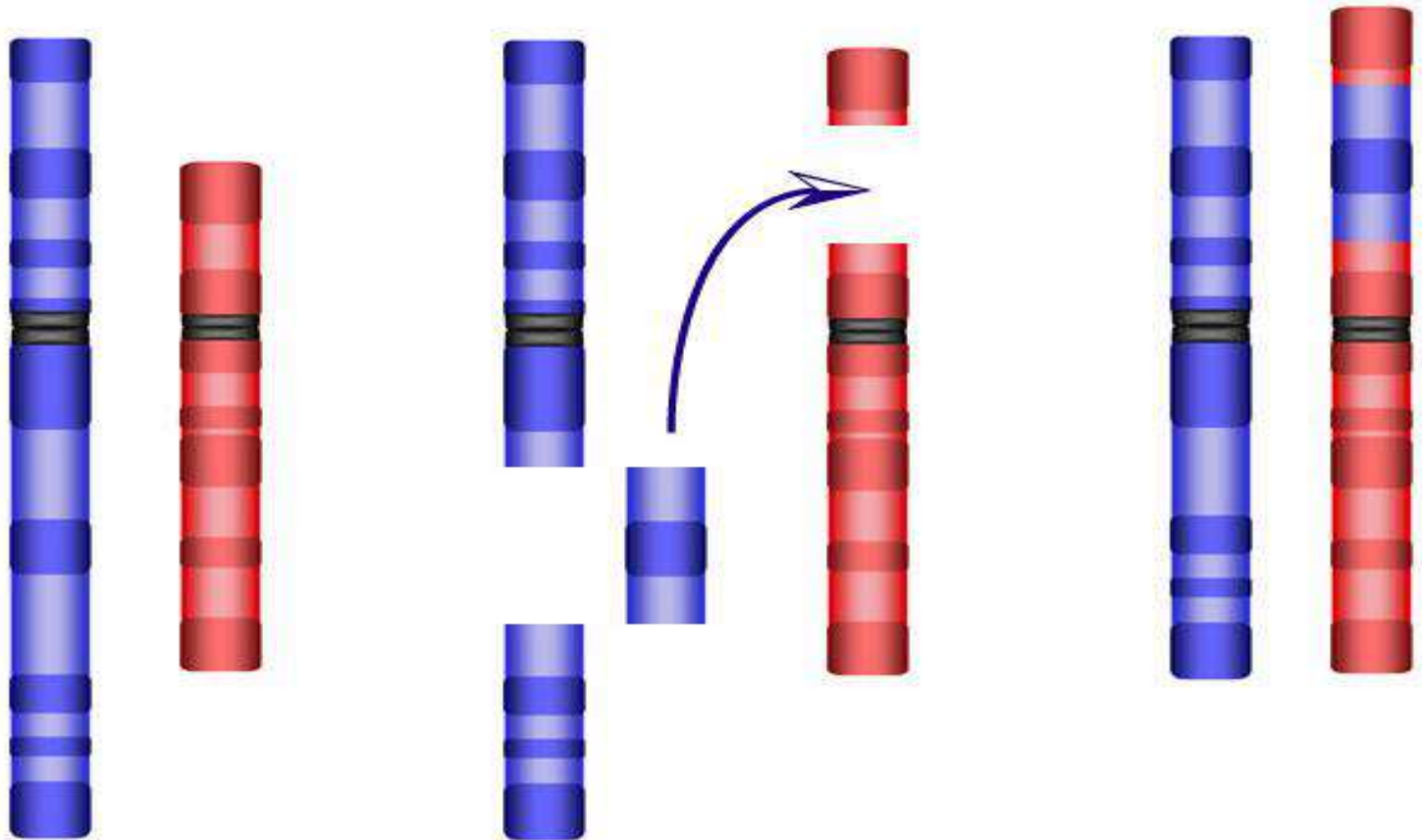


Insertions

A segment removed from one chromosome is inserted into a different chromosome, either in its usual orientation or inverted .

Insertion

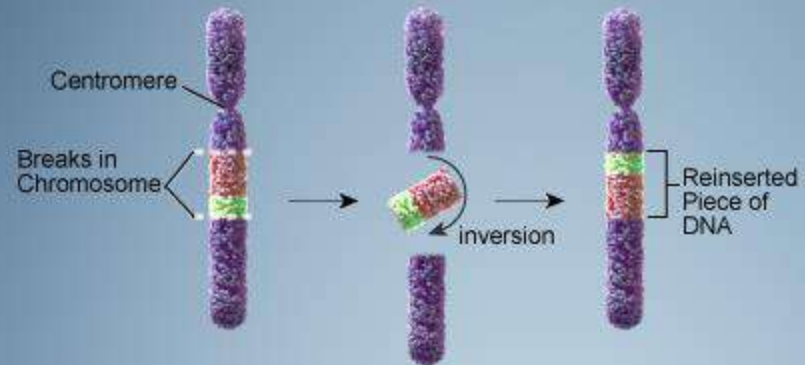
2 chromosomes impliqués
3 points de cassure



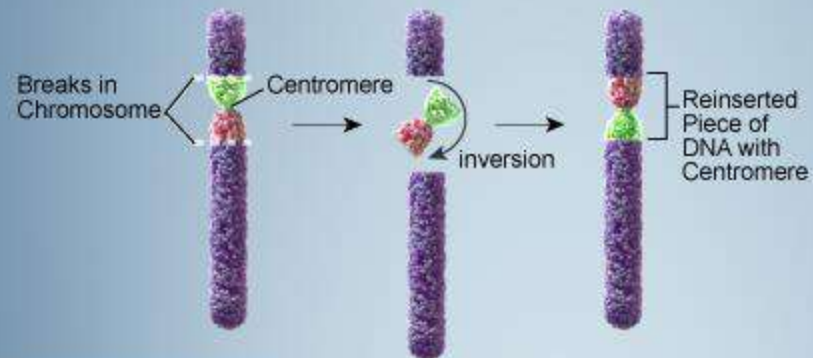
Inversions

- Can occur when a single chromosome undergoes 2 breaks and is reconstituted with the segment between the breaks inverted.
- 2 types:
 - ✓ Paracentric inversions do not include the centromere
 - ✓ pericentric involve the centromere.

Paracentric Inversion



Pericentric Inversion





Inversions

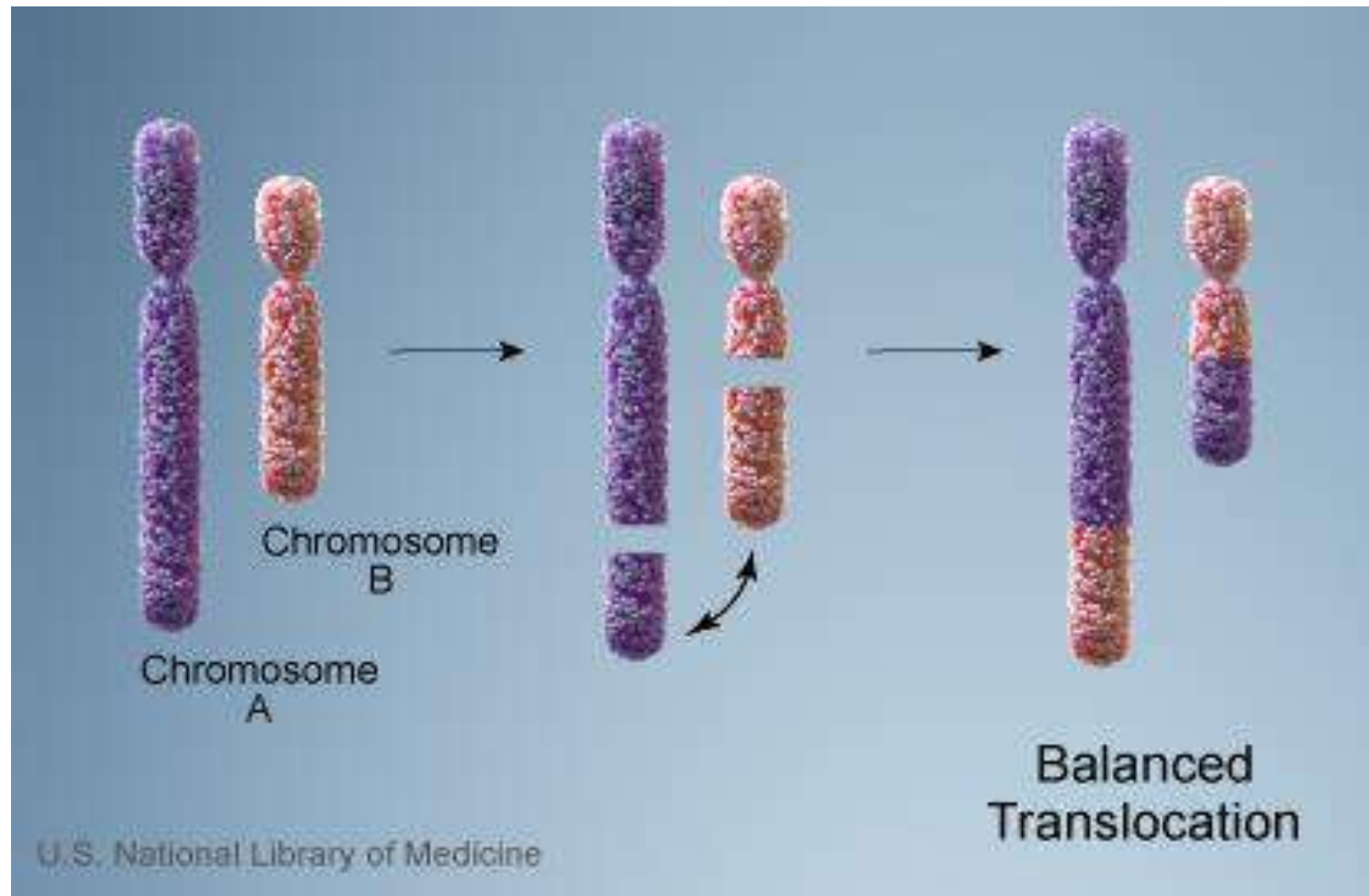
- An inversion doesn't usually cause a phenotypic abnormality in carriers as it is balanced. Its significance is for the progeny, producing abnormal gametes that lead to abnormal offspring.

Translocations

- ❑ Reciprocal translocations (rcp)
- ❑ Robertsonian translocation (rob)

Translocations

- Reciprocal translocations result from breakage of nonhomologous chromosomes with reciprocal exchange of the broken off segment.
- Such translocations are harmless but they are associated with a high risk of unbalanced gametes and abnormal progeny

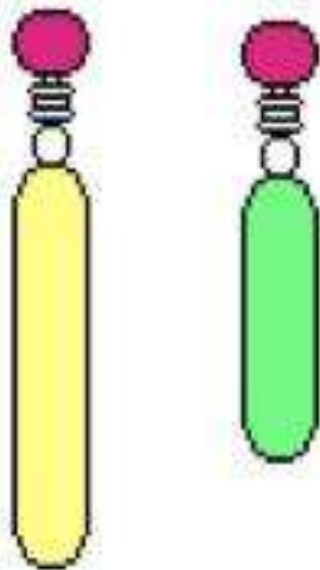


46,XY,t(1;7)(q23.2;q36)



Translocations

- **Robertsonian translocation** involves 2 acrocentric chromosomes that fuse near the centromere and lose their short arms.
- ✓ **Balanced robertsonian translocation:** Chromosomal number is 45 but practically the genome is that of 46 chromosomes.
- ✓ **Unbalanced robertsonian translocation:** chromosomal number is 46 but the genome is that of 47 chromosomes.
- ✓ chromosome which is made of the long arms of the 2 chromosomes is called **derivative chromosome**.



two acrocentric
chromosomes



a Robertsonian
translocation



the short arms
(lost)

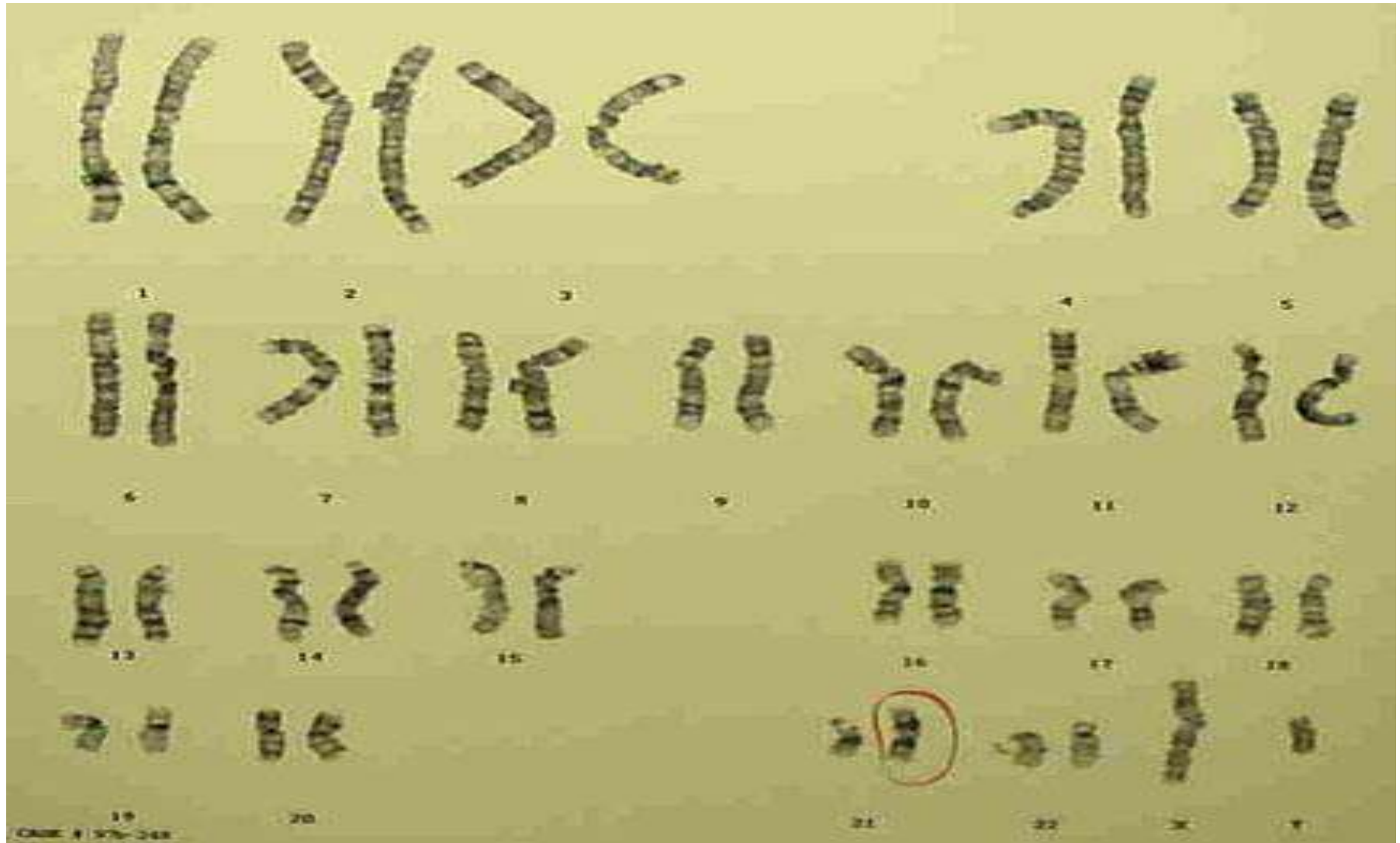
Balanced female carrier: 45,XX,der(21;21)(q10;q10)



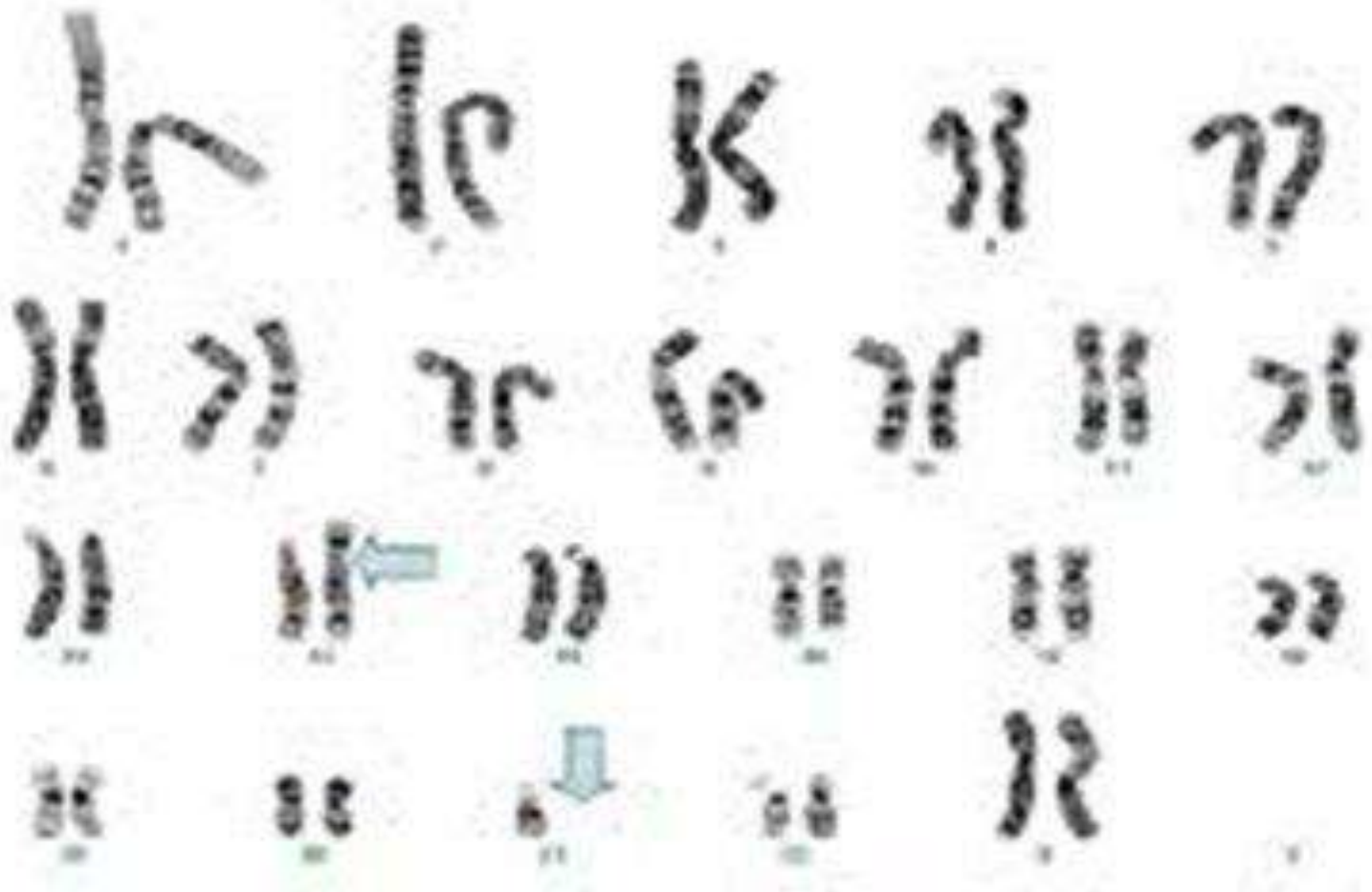
Translocation Down syndrome: 46,XY,der(21;21)

(q10;q10),+21

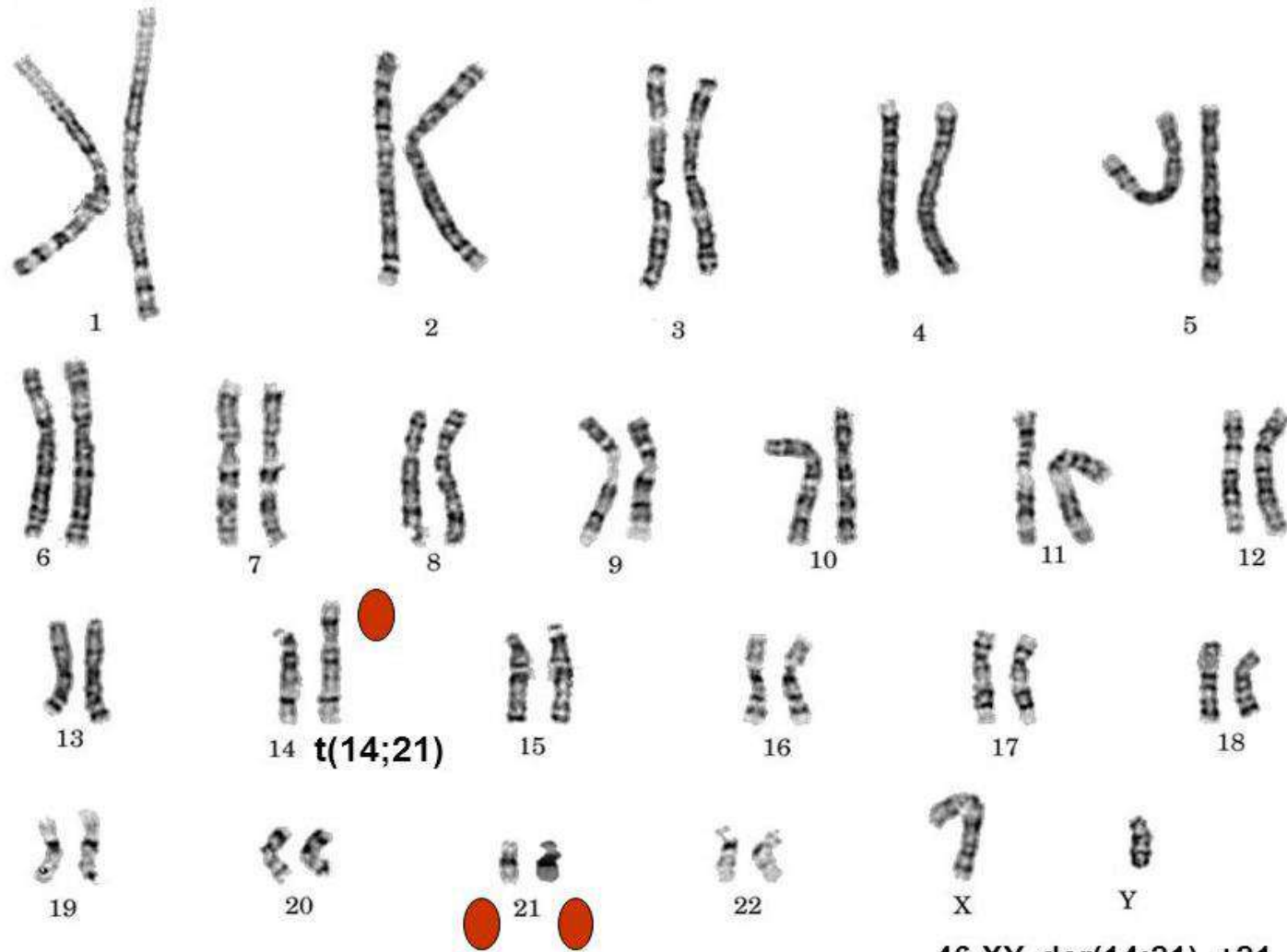
46,XY,rob(21;21)(q10;q10),+21



Balanced female carrier : 45,XX,der(14;21) (q10;q10)/
45,XX,rob (14;21) (q10;q10)/



46,XY,der(14;21) (q10;q10),+21
46,XY,rob(14;21) (q10;q10) ,+21
Translocation Down syndrome



46,XY, der(14;21), +21

✓ Down Syndrome (trisomy 21) due to:

1-Non-disjunction : (47,XX,+21) (47,XY,+21)

2-Translocation

e.g: 46,XX,rob(21;21) (q10;q10), +21

46,XY,rob(14;21) (q10;q10), +21

3-Mosaics

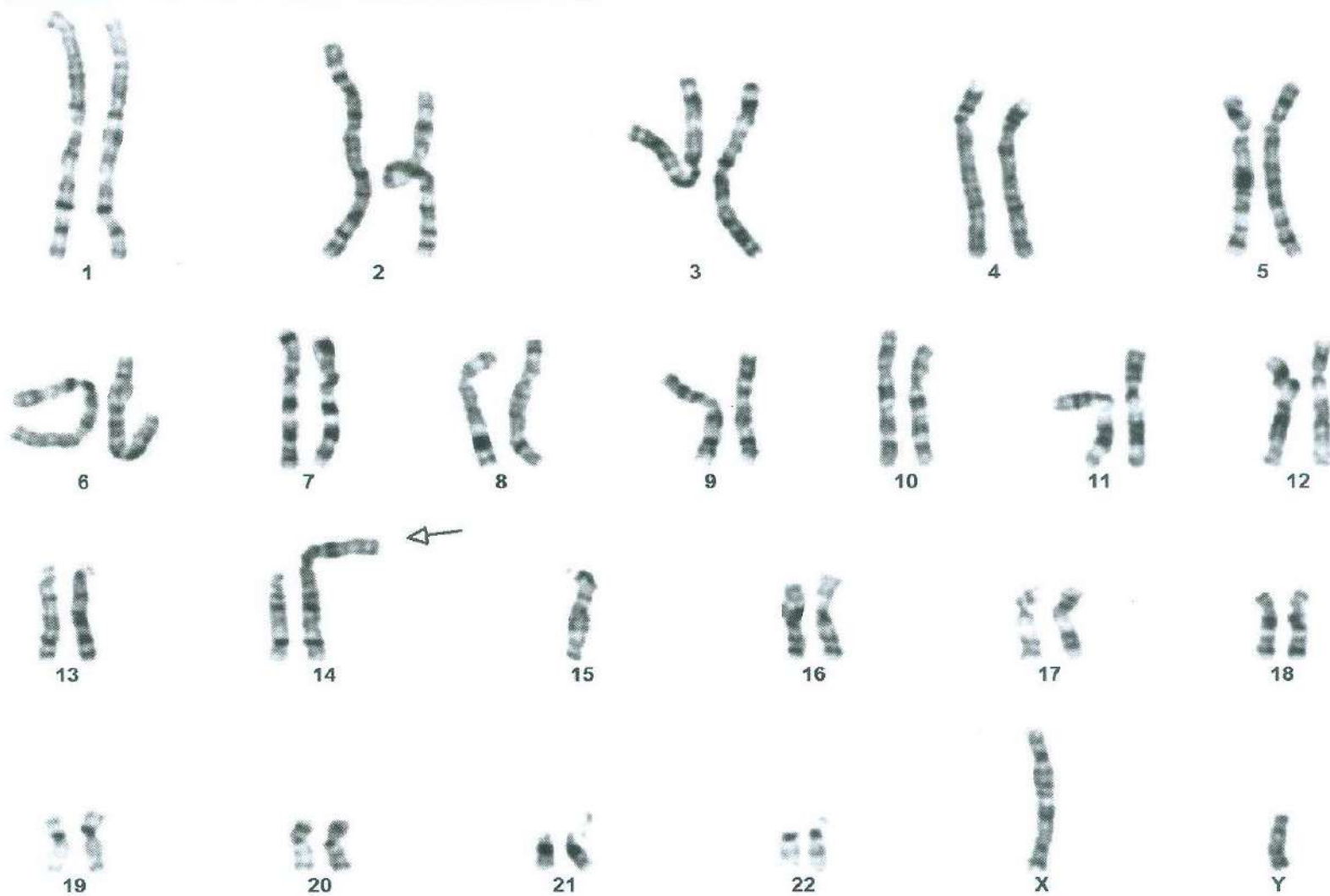
e.g: 47,XX,+21[20]/46,XX[10]

- 
- ✓ Down syndrome : Recurrence Risk for next pregnancy due to non-disjunction : 1% + the maternal age risk

Risks in Robertsonian translocation

- ✓ The female carrier of either a 13q21q or a 14q21q Robertsonian translocation runs a risk of approximately 10-15% for having a baby with Down syndrome, whereas for male carriers the risk is 1% to 3%.
- ✓ In case of a carrier of a 21q21q Robertsonian translocation. All gametes will be either nullisomic or disomic for chromosome 21.
- ✓ Consequently, all pregnancies will end either in spontaneous miscarriage or in the birth of a child with Down syndrome.

45,XY,rob(14;15)(q10;q10)



How to describe chromosomal abnormalities

- ✓ Chromosome report reading 46,XY,t(2;4)(p23;q25) :
Male with a reciprocal translocation between chr 2 and chr 4 involving short arm of chr 2 at region 2 band 3 and long arm of chr 4 at region 2 band 5
- ✓ Chromosome report reading 46,XX,del(15)(q11q13)
Female karyotype with interstitial deletion of chromosome 15 involving region 1 band 1 to region 1 band 3
- ✓ Chromosome report reading 46,XY,del(5)(p15)
Male karyotype with terminal deletion of the short arm of chromosome 5 with the breakage at region 1 band 5

How to describe chromosomal abnormalities

✓ Chromosome report reading 46,XX,inv(9)(p11q13)

Female karyotype with pericentric inversion of chromosome 9 at which breakage and reunion has occurred in short arm region 1 band 1 and long arm region 1 band 3

✓ Chromosome report reading 46,XX,inv(1)(q11q24)

Male karyotype with paracentric inversion of chromosome 1 at which breakage and reunion has occurred in q arm between (involving) region 1 band 1 region 2 band 4

Reference

- ❑ [https:// images.google.com](https://images.google.com)
- ❑ Google Images. The most comprehensive image search on the web.
- ❑ Emery's Elements of Medical Genetics 16th edition 2022 by Turnpenny and Ellard

Thank You!

